



LESSON 5.7

USING THE PYTHAGOREAN THEOREM TO SOLVE REAL-WORLD PROBLEMS

LESSON INTRODUCTION

MA.8.GR.1.1 Apply the Pythagorean theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

LEARNING TARGETS

- I can solve real-world problems by applying the Pythagorean theorem.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

WORKING TOWARD

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean theorem.

ESSENTIAL QUESTIONS

- How can I apply what I have learned about the Pythagorean theorem to real-world problems?
- Why is the Pythagorean theorem useful in real life?

LESSON NARRATIVE

In this lesson, students apply what they learned in Lesson 6. The lesson begins with a collaborative activity where students compare and contrast two routes to determine which one is faster. During 5.7.3 Guided Instruction, the teacher guides students in interpreting a problem and creating a model to represent the problem. Students then try the problem-solving steps on two problems, one where they find the hypotenuse and one where they find the leg. Students are then challenged to apply the Pythagorean theorem to a 3D figure (rectangular prism). Students will have to apply the theorem twice to find the diagonal length. In the independent practice, students solve two problems based on the “I can” statement and one problem based on the “I can” from Lesson 6.

COMPONENT SUMMARY

5.7.1 WARM-UP (~ 5 MIN.)

Activate prior knowledge of Pythagorean theorem

5.7.3 GUIDED INSTRUCTION (15 MIN.)

Pythagorean theorem in context

5.7.5 COLLABORATION (10 MIN.)

Using Pythagorean theorem in 3D figures

5.7.7 WRAP-UP (~5 MIN.)

Practice what was learned this lesson

5.7.2 COLLABORATION (5 MIN.)

Compare and contrast two routes using Pythagorean theorem

5.7.4 GUIDED PRACTICE (5 MIN.)

Practice using Pythagorean theorem in context

5.7.6 YOUR TURN! (5 MIN.)

Practice using Pythagorean theorem in context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Pythagorean theorem
- Hypotenuse

MATERIALS

- (Optional) Highlighters

ADDITIONAL PREPARATION

- Consider posting the steps to finding the missing side of a triangle using the Pythagorean theorem somewhere in your classroom.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget to square each number when using the Pythagorean theorem.
- Students may substitute the numbers in the incorrect spot in the Pythagorean theorem.
- Students may square the numbers incorrectly.
- Students may forget to take the square root at the last step of solving using the Pythagorean theorem.

THINGS TO CONSIDER

- Consider having students write out the formula for the Pythagorean theorem before answering each question. This may help students visualize substituting each number.
- Consider having students color-code the sides of the triangle a , b and c , each corresponding to a certain color.
- In problems that require several steps, students should not round until the final step. Rounding midstep could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values.
- If time is an issue, then consider assigning 5.7.6 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 5.7.2 Collaboration, students need to make an educated guess as to who they think will get there faster. Prompt students with questions like: What do you notice about the model? What do you wonder? What do you think you would need to find out in order to answer this question?

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In the previous lesson, students practiced applying the Pythagorean theorem without context. In this lesson, students apply what they have learned within the context of a real-world problem. Students are connecting the mathematics to a real-world context while using models to represent and solve problems.

DIFFERENTIATE INSTRUCTION

In 5.7.2 Collaboration, provide a kinesthetic approach to this problem. Have two students stand in the corner of the room. One student will walk diagonally across the room, and the other student will walk the perimeter of the room. Who gets to the opposite corner first? Does their speed matter? Engaging students in this activity will connect the content on the paper to real-world application, deepening understanding and increasing engagement in what can sometimes be an abstract concept.

5.7.1 WARM-UP: BELL WORK

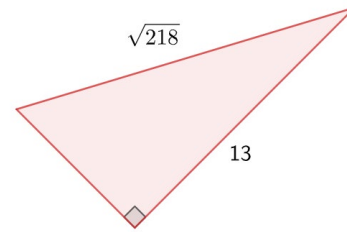
TEACHER MOVES

This error analysis contains a common misconception in student work. Students may make this same mistake of substituting the value into the wrong spot. Take a few minutes to talk about strategies to help substituting into the correct spot in the Pythagorean Theorem.



5.7.1 Warm-Up: Bell Work

Sawyer applied the Pythagorean Theorem to find the length of the missing side of the triangle. His work is shown.



$$\begin{aligned} a^2 + b^2 &= c^2 \\ 13^2 + \sqrt{218}^2 &= c^2 \\ 169 + 218 &= c^2 \\ 387 &= c^2 \\ \sqrt{387} &= c \end{aligned}$$

The missing side is $\sqrt{387}$ units long.

- Write a note to Sawyer explaining his mistake.
Answers will vary. You made a mistake when substituting the value of $\sqrt{218}$ for b^2 . The value of $\sqrt{218}$ should have been substituted for c^2 .

- Correct Sawyer's mistake.

$$\begin{aligned} 13^2 + b^2 &= \sqrt{218}^2 \\ 169 + b^2 &= 218 \\ b^2 &= 49 \\ b &= 7 \end{aligned}$$

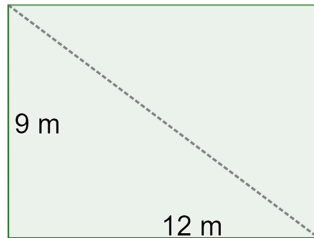


5.7.2 Collaboration: Who Wins?

Mai and Tyler were standing at one corner of a large rectangular field and decided to race to the opposite corner. Since Mai had a bike and Tyler did not, they thought it would be a more equal race if Mai rode along the sidewalk that surrounds the field while Tyler ran the shorter distance directly across the field.

The field is 12 meters (m) long and 9 m wide.

1. Tyler can run at around 5 m per second, and Mai can ride her bike at around 7.5 m per second.



- a. Before making any calculations, predict who you think will win and by how much. Show or explain your thinking.

Answers will vary.

- b. Work with your partner to mathematically determine who will win the race.

Mai wins by 0.2 seconds.

$$\text{Tyler: } \frac{15}{5} = 3 \text{ sec}$$

$$\text{Mai: } \frac{21}{7.5} = 2.8 \text{ sec}$$

5.7.2 COLLABORATION: WHO WINS?

TEACHER MOVES

The lesson provides many different opportunities for students to solidify their understanding of the Pythagorean theorem by using it to solve problems. If your students are struggling, then consider using this component as a challenge.

DIFFERENTIATE INSTRUCTION

For a kinesthetic approach to this problem, have two students stand in the corner of the room. One student will walk diagonally across the room, and the other student will walk the perimeter of the room. Who gets to the opposite corner first? Does their speed matter?

ENRICHMENT

Students may struggle to determine who will win the race because it requires several steps. Students must first figure out the length of the hypotenuse, then figure out the length of the sum of the sides, then divide by the respective speeds to find the winner. Scaffold through questioning.

- What information do you have already?
- What information do you need to find out who will win the race?
- How can you find the length Tyler will run?
- What do you need to do to determine how long it will take him to run that distance?

5.7.3 GUIDED INSTRUCTION: APPLYING THE PYTHAGOREAN THEOREM

DIFFERENTIATE INSTRUCTION

Consider creating an anchor chart and posting these steps in your classroom somewhere for students to reference.

TEACHER MOVES

Common Misconception

Students may mislabel the diagram of the problem. Check for students' understanding of which information should go where. Prompt students with questions like:

- What information are we given?
- Is 12 a leg or the hypotenuse? How do you know?

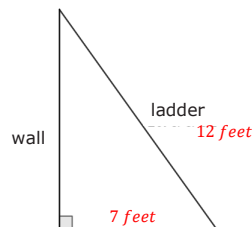


5.7.3 Guided Instruction: Applying the Pythagorean Theorem

The Pythagorean Theorem can be used to find an unknown side length of any right triangle.

Pythagorean Theorem Problem-Solving Steps

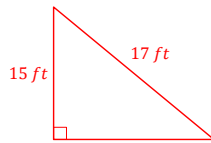
1. Model the problem with a diagram (if not provided).
 2. State the Pythagorean Theorem.
 3. Substitute given values.
 4. Solve for the unknown.
1. A 12-foot long ladder leans against a vertical wall. The base of the ladder is 7 feet from the bottom of the wall. Determine, to the nearest tenth of a foot, how high the ladder reaches up the wall.



- a. Label the diagram with the given information.
- b. State Pythagorean Theorem $a^2 + b^2 = c^2$
- c. Substitute given values $7^2 + b^2 = 12^2$
- d. Solve for the unknown $b \approx 9.7 \text{ feet}$

2. A cable is being placed on level ground to support a tower. The cable is 17 feet (ft) and should be connected 15 ft up the tower. How far away from the bottom of the tower should the other end of the cable connect to the ground?

a. Model the problem.



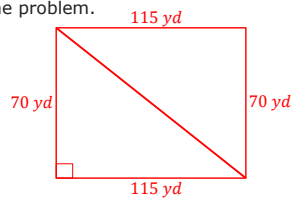
b. State Pythagorean Theorem $a^2 + b^2 = c^2$

c. Substitute given values $a^2 + 15^2 = 17^2$

d. Solve for the unknown. $a = 8 \text{ feet}$

3. A soccer field is 115 yards (yd) long by 70 yd wide. What is the length of the diagonal of a soccer field? Round your answer to the nearest tenth.

a. Model the problem.



b. State Pythagorean Theorem $a^2 + b^2 = c^2$

c. Substitute given values $70^2 + 115^2 = c^2$

d. Solve for the unknown $c \approx 134.6 \text{ yd}$

5.7.3 GUIDED INSTRUCTION: APPLYING THE PYTHAGOREAN THEOREM (CONTINUED)

DIFFERENTIATE INSTRUCTION

Prompt students to highlight important information. Students can color-code the highlighted information in the situation, the diagram, and the equation.

ENRICHMENT

Students may struggle to model and label the problem accurately. Help students through questioning.

- What does 15 ft represent in the problem? Where should you show this on the triangle? Why?
- What does 17 ft represent in the problem? What does it represent on the triangle? How do you know?
- How can you label the diagram to help you substitute the information in the equation correctly?

DIFFERENTIATE INSTRUCTION

Consider providing pre-drawn models for students who might have a hard time creating them on their own. Then students just have to label the models accurately, providing an additional entry point for learners.

5.7.4 GUIDED PRACTICE: APPLYING THE PYTHAGOREAN THEOREM

DIFFERENTIATE INSTRUCTION

Encourage students to create pictures or diagrams as they complete the questions in this section. Illustrating what is happening in the problem will provide a visual scaffold as students work to determine the unknown lengths.



5.7.4 Guided Practice: Applying the Pythagorean Theorem

Solve each problem using the Pythagorean Theorem.

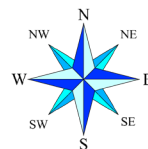
1. Malik leaves his house and drives 7 miles (mi) west. He then turns and drives 5 mi north. How far away is he from his house? Round your answer to the nearest tenth.

$$5^2 + 7^2 = c^2$$

$$25 + 49 = c^2$$

$$74 = c^2$$

$$c \approx 8.6 \text{ miles}$$



2. Audrey designs a 15-foot slide from her bedroom window into her swimming pool. The end of the slide is 9 feet from the base of her house. How high up is her window? Round your answer to the nearest tenth.

$$a^2 + 9^2 = 15^2$$

$$a^2 + 81 = 225$$

$$a^2 = 144$$

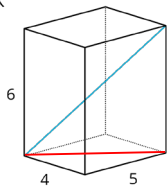
$$a = 12 \text{ ft.}$$



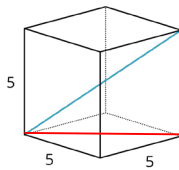
5.7.5 Collaboration: Internal Dimensions

1. Two rectangular prisms are shown:

K



L



- Predict which figure has a longer diagonal. Note that the figures are not drawn to scale.
Figure K
- Discuss with your partner if there is enough information to find the diagonal using the Pythagorean Theorem. Summarize your discussion.
Answers will vary. There is not enough information because the diagonal of the base of each prism is needed and it is not given. There is enough information because the values needed for both legs are given and the hypotenuse needed can be found.
- What other measurement must be calculated first to find the diagonal? Label this extra measurement on the figures above.
The diagonal of the base of each prism is needed.
- Find the exact length of the base diagonals using the Pythagorean Theorem.
 $\sqrt{41}$, $\sqrt{50}$
- Use the Pythagorean Theorem to calculate the length of both diagonals. Which one is longer?
Figure K

5.7.5 COLLABORATION: INTERNAL DIMENSIONS

TEACHER MOVES

The lesson provides many different opportunities for students to solidify their understanding of the Pythagorean theorem by using it to solve problems. If your students are struggling, then consider using this component as a challenge.

With multistep problems like this, reinforce problem-solving strategies that are used in your classroom. Make note that to solve this problem, students will have to apply the Pythagorean theorem twice.

ENRICHMENT

Help guide student discussions by asking guiding questions.

- What information do we need to solve this problem?
- How many right triangles do you see?
- How can you use the information to find the missing diagonals?

DIFFERENTIATE INSTRUCTION

Encourage students who are struggling to use highlighters or colored pencils to label the different parts of the rectangular prisms.

5.7.6 YOUR TURN!

TEACHER MOVES

Use observational data from the 5.7.3 Guided Instruction and 5.7.4 Guided Practice to determine how much guidance is needed to bring together key ideas in this Your Turn! component. If necessary, consider modeling one of the questions to address common misconceptions. Alternatively, use the observational data to work with a small group of students on these three problems.

TEACHER MOVES

Students may have a hard time visualizing this situation. Consider helping create a diagram to model this situation as a class.

- What information is given?
- What do we need to know?
- What would be the longest distance across the opening of the trash can?

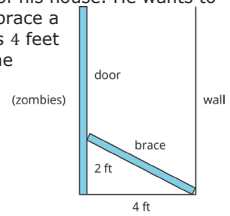


5.7.6 Your Turn!

Solve each problem by applying the Pythagorean Theorem.

1. A man is trying to zombie-proof his house. He wants to cut a length of wood that will brace a door against a wall. The wall is 4 feet (ft) away from the door, and he wants the brace to rest 2 ft up the door. About how long should he cut the brace?

4.5 ft



2. Lovelace Park in Jacksonville is a rectangular-shaped park, measuring approximately 410 meters (m) by 360 m. Joseirie wants to get from one corner of the park to the opposite corner. She wonders about the difference between cutting across the diagonal through the park compared to going around the park, along the streets.

How much shorter would her walk be cutting through the park? Round your answer to the nearest tenth.

224.4 m

3. At a restaurant, a trash can's opening is rectangular and measures 7 inches (in) by 9 in. The restaurant serves food on trays that measure 12 in by 16 in. Jada says it is impossible for the tray to accidentally fall through the trash can opening because the shortest side of the tray is longer than either edge of the opening.

Do you agree or disagree with Jada's explanation? Justify mathematically.

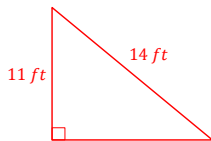
Answers will vary. I disagree with Jada's justification. The diagonal distance of the opening is $7^2 + 9^2 = c^2$, or ≈ 11.4 in. Even the shorter side of the tray is too wide for the trash can opening.

**5.7.7 Wrap-Up: Ticket Out the Door**

- Jonathan places a 14-foot ladder against the side of the school. The top of the ladder is 11 feet above the ground.

How far from the building is the base of the ladder?
Round your answer to the nearest tenth.

8.7 ft

**5.7.7 WRAP-UP: TICKET OUT THE DOOR****DIFFERENTIATE INSTRUCTION**

This question allows students to show understanding of finding the missing side using the Pythagorean theorem in context. The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.